

The Planck-Scale Structure of the Conversion Factors  
Why  $G = (\ell_P^2 \times c^3)/\hbar$  justifies the form of the factors from  
Document 012 T0 Theory: From Dimensionless to SI

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### Correction notes

**Corrections from the calculation review (30 Sept. 2026).** The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Numerical verification, path from T0:  $G_{\text{nat}} \times C_{\text{conv}} \times K_{\text{frak}}$  with  $3.1 \times 10^{-10} \times 7.8 \times 10^{-3} \times 0.986 \times 10^1$  (gives  $2.38 \times 10^{-11}$ )  $\rightarrow \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \approx 6.69 \times 10^{-11}$ , without  $C_{\text{dim}}$ .
- SI conversion factor: " $C_{\text{conv}} \sim c^3/\hbar$  (order of magnitude)"  $\rightarrow$  structure only; numerically  $c^3/\hbar = 2.6 \times 10^{59}$ , and  $C_{\text{conv}} = 7.8 \times 10^{-3}$  is the SI conversion of  $G$ .
- Code excerpt from calc\_De.py: note,  $G_{\text{nat}} \times 2.843 \times 10^{-5} \approx 8.7 \times 10^{-15}$ ;  $2.843 \times 10^{-5} \neq C_{\text{conv}} K_{\text{frak}}$ .
- Path via the Planck formula: checks the dimension, not the numerical value, since  $\ell_P$  is defined via  $G$  (Doc. 012).

## Abstract

This document explains why the conversion factors in Document 012 have the exact form they do. The mathematical relation  $G = \frac{\ell_P^2 \times c^3}{\hbar}$  is not a new calculation method (it is a rearrangement of the known Planck length definition), but it reveals the *fundamental structure* underlying the conversion factors.

**Core Message:** The factors  $C_{\text{dim}}$ ,  $C_{\text{conv}}$ , and  $K_{\text{frak}}$  in Document 012 are not arbitrary but follow from the Planck-scale structure of  $G$ . The formula also serves as a consistency check: if all factors are correct,  $G_{\text{SI}} = \frac{\ell_P^2 \times c^3}{\hbar}$  must be satisfied.

**For the complete technical derivation** of all conversion factors, see Document 012.

## 1 The Problem: Conversion from T0 to SI

### Recap: The T0 Formula for G

From Document 012 it is known:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}} \quad (1)$$

**With the factors:**

- $\frac{\xi^2}{4m_e} \approx 8.7 \times 10^{-9} \text{ MeV}^{-1}$  (from T0 geometry)
- $C_{\text{dim}} \approx 3.5 \times 10^{-2}$  (dimension correction)
- $C_{\text{conv}} \approx 7.8 \times 10^{-3} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \cdot \text{MeV}$  (SI conversion)
- $K_{\text{frak}} = 0.986$  (fractal correction)

### The Question

**Why do these factors have exactly this form?**

Specifically:

- Why does  $c^3$  appear? (in  $C_{\text{conv}}$ )
- Why  $\hbar$  in the denominator?

- Why a length scale squared?
- What is the fundamental structure?

## .2 The Planck Length as the Starting Point

### Standard Definition (since Max Planck, 1899)

The Planck length is defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (2)$$

#### Standard Interpretation:

- $G$  is a fundamental constant (measured)
- $\ell_P$  is calculated from it
- $\ell_P \approx 1.616 \times 10^{-35}$  m
- Quantum gravity scale

### Mathematical Rearrangement

From  $\ell_P = \sqrt{\frac{\hbar G}{c^3}}$  it follows by rearrangement:

$$\ell_P^2 = \frac{\hbar G}{c^3} \quad (3)$$

$$\ell_P^2 \times c^3 = \hbar G \quad (4)$$

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad (5)$$

This rearrangement is the structure to which the following sections refer.

## .3 The Structure of the Conversion Factors

### What Does the Planck Formula Show?

[Fundamental Structure]

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad (6)$$

**Dimensional Analysis:**

$$[G] = \frac{[\ell_P^2] \times [c^3]}{[\hbar]} \quad (7)$$

$$= \frac{[\text{m}^2] \times [\text{m}^3/\text{s}^3]}{[\text{J} \cdot \text{s}]} \quad (8)$$

$$= \frac{[\text{m}^5/\text{s}^3]}{[\text{kg} \cdot \text{m}^2/\text{s}^2 \cdot \text{s}]} \quad (9)$$

$$= \frac{[\text{m}^5/\text{s}^3]}{[\text{kg} \cdot \text{m}^2/\text{s}]} \quad (10)$$

$$= \frac{[\text{m}^3]}{[\text{kg} \cdot \text{s}^2]} \quad (11)$$

This yields the dimension  $[G] = \text{m}^3/(\text{kg} \cdot \text{s}^2)$  (dimension check satisfied).

**Connection to T0 Factors**

In T0, one starts with  $G_{\text{nat}}$  in dimension  $[E^{-2}]$  (Energy<sup>-2</sup>).

**Conversion  $[E^{-2}] \rightarrow [\text{m}^3/(\text{kg} \cdot \text{s}^2)]$  must have:**

$$[E^{-2}] \times \text{Factor} = [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \quad (12)$$

**The factor must have the structure:**

$$\text{Factor} = \frac{[\text{Length}^3]}{[\text{Energy}]} \quad (13)$$

**From the Planck formula:**

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \Rightarrow \text{Structure: } \frac{[\text{Length}^2] \times [\text{Velocity}^3]}{[\text{Action}]} \quad (14)$$

With  $[\hbar] = [\text{Energy} \times \text{Time}]$  and  $[c] = [\text{Length}/\text{Time}]$ :

$$\frac{[\ell_P^2 \times c^3]}{[\hbar]} = \frac{[\text{Length}^2] \times [\text{Length}^3/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \quad (15)$$

$$= \frac{[\text{Length}^5/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \quad (16)$$

$$= \frac{[\text{Length}^5]}{[\text{Energy} \times \text{Time}^4]} \quad (17)$$

**This justifies why:**

- $c^3$  in the numerator ( $\text{Length}^3/\text{Time}^3$ )
- $\hbar$  in the denominator ( $\text{Energy} \times \text{Time}$ )
- $\text{Length}^2$  (from  $\ell_P^2$ )
- The combination yields  $[\text{m}^3/(\text{kg} \cdot \text{s}^2)]$

## .4 Justification of the Factors in Document 012

### The Dimension Correction Factor $C_{\text{dim}}$

From Document 012:

$$C_{\text{dim}} = \frac{1}{E_{\text{char}}} \approx 3.5 \times 10^{-2} \quad [\text{MeV}^{-1}] \quad (18)$$

With  $E_{\text{char}} = 28.4 \text{ MeV}$  (7-step derivation in Doc. 012).

#### Why this factor?

The T0 formula  $G = \frac{\xi^2}{4m_e}$  initially yields dimension  $[E^{-1}]$ .

In natural units, however,  $G$  requires  $[E^{-2}]$ .

$\Rightarrow$  Factor  $[E^{-1}]$  needed:  $C_{\text{dim}} = 1/E_{\text{char}}$

#### Connection to the Planck structure:

The energy scale  $E_{\text{char}}$  is not arbitrary but emerges from the same geometry as  $\ell_P$ . It is the characteristic scale where T0 geometry connects to the Planck scale.

### The SI Conversion Factor $C_{\text{conv}}$

From Document 012:

$$C_{\text{conv}} \approx 7.8 \times 10^{-3} \quad [\text{m}^3 \text{kg}^{-1} \text{s}^{-2} \cdot \text{MeV}] \quad (19)$$

#### Structure of this factor:

$$C_{\text{conv}} \sim \frac{c^3}{\hbar} \quad (\text{in appropriate units}) \quad (20)$$

$$= \frac{(2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \approx 2.6 \times 10^{59} \quad (\text{SI numerical value, structure only}) \quad (21)$$

(Corrected on 30 Sept. 2026: previously "(order of magnitude)" – in SI,  $c^3/\hbar = 2.6 \times 10^{59}$  does not correspond numerically to  $C_{\text{conv}} = 7.8 \times 10^{-3}$ ;  $C_{\text{conv}}$  is the SI conversion of  $G$ , and the relation to  $c^3/\hbar$  concerns only the structure.)

#### Why exactly this combination?

The Planck formula  $G = \frac{\ell_P^2 c^3}{\hbar}$  shows:

- $c^3$  converts time scale to space scale (dimension:  $\text{m}^3/\text{s}^3$ )
- $\hbar$  connects energy with frequency (dimension:  $\text{J} \cdot \text{s}$ )
- Combination  $c^3/\hbar$  has dimension  $[\text{m}^3/(\text{kg} \cdot \text{s}^2)]/[\text{Energy}]$

This is the role played by  $C_{\text{conv}}$ .

## Numerical Verification

### Consistency Check

From T0 (Document 012):

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \approx 3.1 \times 10^{-10} \quad [E^{-2}] \quad (22)$$

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (23)$$

$$\approx 8.7 \times 10^{-9} \times 7.8 \times 10^{-3} \times 0.986 \quad (24)$$

$$\approx 6.69 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \quad (25)$$

(Corrected on 30 Sept. 2026: previously  $G_{\text{SI}} = G_{\text{nat}} \times C_{\text{conv}} \times K_{\text{frak}} \approx 3.1 \times 10^{-10} \times 7.8 \times 10^{-3} \times 0.986 \times 10^1 \approx 6.67 \times 10^{-11}$  – this product gives  $2.38 \times 10^{-11}$ , and the factor  $10^1$  has no origin; only  $\frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$  without  $C_{\text{dim}}$  is consistent.)

From Planck Formula (Verification):

$$\ell_P = 1.616 \times 10^{-35} \text{ m} \quad (26)$$

$$c = 2.998 \times 10^8 \text{ m/s} \quad (27)$$

$$\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s} \quad (28)$$

$$G_{\text{check}} = \frac{\ell_P^2 \times c^3}{\hbar} \quad (29)$$

$$= \frac{(1.616 \times 10^{-35})^2 \times (2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \quad (30)$$

$$= \frac{2.611 \times 10^{-70} \times 2.694 \times 10^{25}}{1.055 \times 10^{-34}} \quad (31)$$

$$= 6.67 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \quad (32)$$

Both paths give  $6.7 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ . The second path checks the dimension, not the numerical value, since  $\ell_P$  is defined via  $G$  (see Doc. 012). (Corrected on 30 Sept. 2026: previously "Both paths give  $6.67 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$  to the digits shown." – with the rounded factors the first path gives  $6.69 \times 10^{-11}$ ; the Planck formula only returns  $G$  because  $\ell_P$  is computed from  $G$ .)

**This shows:** The factors in Doc. 012 have the dimensionally appropriate structure.

## .5 The Role of the Planck Formula in T0

### Not Circular in T0

Why is the formula not circular?

**Standard Physics (circular):**

1. Measure  $G$
2. Calculate  $\ell_P = \sqrt{\hbar G / c^3}$
3. Calculate  $G = \ell_P^2 c^3 / \hbar$   
 $\Rightarrow$  Get  $G$  back (no information gained)

**T0 Physics (not circular):**

1. Determine  $\xi$  from experiment (via  $\alpha$ ,  $E_0$ )

2. Calculate  $G_{SI}$  from  $\xi$  (with factors)
3. Calculate  $\ell_P = \sqrt{\hbar G_{SI}/c^3}$
4. Check:  $G_{SI} = \ell_P^2 c^3 / \hbar$   
 $\Rightarrow$  Consistency check

### Three Uses of the Planck Formula

1. **Justification:** Shows why factors have the form  $c^3/\hbar$  etc.
2. **Verification:** Consistency check for calculated  $G$
3. **Structural Insight:**  $G$  emerges at the Planck scale

## .6 Practical Application: Python Implementation

### Code Structure (from calc\_De.py)

The T0 calculation script shows exactly this logic:

```
# Main calculation (from ξ)
G_t0_dimensionless = (xi**2) / (4 * m_char)
conversion_factor_nat = 3.521e-2 # C_dim
G_nat = G_t0_dimensionless * conversion_factor_nat

SI_conversion_factor = 2.843e-5 # C_conv × K_frak
G_SI = G_nat * SI_conversion_factor

# Planck formula as verification
planck_conversion_factor = (l_P**2 * c**3) / hbar

# Check: Both should agree
assert abs(G_SI - planck_conversion_factor) < 1e-13
```

*(Note of 30 Sept. 2026: with the numbers of this code excerpt,  $G_{nat} \times 2.843 \times 10^{-5} \approx 8.7 \times 10^{-15}$ , not  $G$ ; moreover,  $2.843 \times 10^{-5}$  is not  $C_{conv} \times K_{frak} \approx 7.7 \times 10^{-3}$ . Consistent is  $\frac{\xi^2}{4m_e} C_{conv} K_{frak}$  without  $C_{dim}$ , see the numerical verification.)*

### What the Code Shows

- **Lines 1-2:** T0 formula  $\xi^2/(4m)$
- **Line 3:** Dimension correction  $C_{dim}$  (corresponds to  $1/E_{char}$ )
- **Line 5:** SI conversion  $C_{conv} \times K_{frak}$  (corresponds to  $c^3/\hbar$  structure)
- **Line 8:** Planck formula for verification
- **Line 11:** Both paths must agree.

## .7 Comparison with Electrodynamics

### Analogy: Speed of Light

In electrodynamics:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (33)$$

#### Interpretation:

- $c$  emerges from electromagnetic vacuum structure
- $\mu_0, \epsilon_0$  describe vacuum properties
- Formula shows structure, not calculation

### Analogy: Gravitational Constant

In T0:

$$G = \frac{\ell_p^2 \times c^3}{\hbar} \quad (34)$$

#### Interpretation:

- $G$  emerges from spacetime geometry (T0)
- $\ell_p, c, \hbar$  describe geometry properties
- Formula shows structure, justifies conversion factors

### Parallelism

| Aspect       | Electrodynamics                 | Gravitation                 |
|--------------|---------------------------------|-----------------------------|
| Constant     | $c$                             | $G$                         |
| Formula      | $c = 1/\sqrt{\mu_0 \epsilon_0}$ | $G = \ell_p^2 c^3 / \hbar$  |
| Emerges from | EM vacuum                       | Spacetime geometry          |
| Justifies    | $\mu_0, \epsilon_0$ structure   | $C_{\text{conv}}$ structure |

**Table 1:** Parallel Structures

## .8 Summary

### The Central Message

#### Structure Justification

**The Planck formula  $G = \frac{\ell_p^2 \times c^3}{\hbar}$  is useful for T0 because it:**

1. **Justifies** why the conversion factors in Doc. 012 have exactly the form:
  - $C_{\text{dim}} \sim 1/E$  (energy scale)
  - $C_{\text{conv}} \sim c^3/\hbar$  (Planck structure)
2. **Serves as a consistency check:**
  - Calculate  $G$  from  $\xi$  with factors
  - Calculate  $\ell_p$  from  $G$

- Check:  $G = \ell_p^2 c^3 / \hbar$

### 3. Shows the geometric structure:

- $G$  emerges at Planck scale  $\ell_p$
- Connection quantum mechanics ( $\hbar$ )  $\leftrightarrow$  relativity ( $c$ )
- Fundamental role of geometry

**It is not a new calculation method (that would be circular), but it justifies the structure of the factors.**

## What is New?

### Mathematically NOT new:

- The formula  $G = \ell_p^2 c^3 / \hbar$  (rearrangement of  $\ell_p$  definition since 1899)
- The Planck units (Max Planck, 1899)

#### New in T0:

- The formula *justifies* the conversion factors
- It serves as *verification* (not circular, since  $G$  comes from  $\xi$ )
- It shows that  $G$  emerges at the Planck scale
- $\ell_p$  is not fundamental but follows from  $G$  (which follows from  $\xi$ )

## Connection to Document 012

**Document 012 shows:** HOW to calculate  $G$  from  $\xi$  (all steps)

**This document (127) shows:** WHY the factors have this structure

**Together:** Calculation and structural justification of  $G$  in T0

## Practical Significance

### For calculations:

- Use the T0 path:  $\xi \rightarrow G$  (Doc. 012)
- Planck formula as a check
- Both must agree

### For understanding:

- Planck formula shows structure
- Justifies why  $c^3/\hbar$  appears
- Shows geometric origin

### For philosophy:

- $G$  is not fundamental
- $G$  emerges at the Planck scale
- $G$  follows from  $\xi$  together with the conversion factors of Doc. 012